

## CMPSC445 Project 2

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### Best Restaurants in Pennsylvania

#### 1. Description of the Project

The objective of this project is to design and implement a full-stack web application that identifies and ranks the best restaurants in Pennsylvania using a data-driven approach. The system enables users to explore restaurant rankings by selecting a specific city and viewing a dynamically generated list of top-performing establishments. The ranking is computed using key business attributes such as average star rating and review count, which serve as indicators of customer satisfaction and popularity.

This project integrates multiple components learned throughout the course, including data preprocessing, feature engineering, model design, and web application development. The backend processes the dataset and computes ranking scores, while the frontend provides an interactive interface for users to query and visualize results. The overall goal is to demonstrate how machine learning concepts can be applied to a practical and meaningful real-world problem.

#### 2. Significance of the Project

This project is significant because it addresses a common real-world decision-making problem: choosing where to eat. Many existing platforms rely solely on average ratings, which can be misleading when not considering the number of reviews or overall popularity. By combining multiple factors into a single scoring function, this project provides a more balanced and reliable ranking of restaurants.

The novelty of the project lies in its ability to transform raw business data into an interpretable “Best in PA” scoring system. Instead of presenting static data, the application allows users to interact with the system and retrieve customized results based on location. Additionally, the integration of a ranking model within a web application demonstrates how machine learning techniques can be deployed to support user-centric decision-making. This makes the project both technically meaningful and practically useful.

#### 3. Data Collection

The dataset used in this project was obtained from Kaggle and is based on Yelp-style business data. It contains structured information about restaurants, including their names, locations,

ratings, and review statistics. The dataset provides a realistic representation of business performance and customer engagement, making it suitable for ranking and analysis tasks.

To align with the project theme, the dataset was filtered to include only businesses located in Pennsylvania (state = PA). This ensures that all results are relevant to the “Best in PA” objective. The dataset contains thousands of entries, with each row representing a unique restaurant along with its associated metadata.

Dataset Features (Metadata)

- business\_id: Unique identifier for each business
- name: Restaurant name
- city: Location of the restaurant
- state: State abbreviation (PA)
- stars: Average customer rating (scale of 1–5)
- review\_count: Total number of reviews
- categories: Type of business (e.g., restaurants, cafes)
- WiFi\_status: Does business have free accessible wifi.

Sample Data

| Name               | City         | Stars | Review Count |
|--------------------|--------------|-------|--------------|
| St Honore Pastries | Philadelphia | 4.0   | 80           |
| Example Cafe       | Pittsburgh   | 4.5   | 120          |

This dataset provides both quality indicators (ratings) and popularity indicators (review counts), which are essential for building an effective ranking system.

4. Data Processing and Feature Engineering

Before building the ranking model, the dataset underwent several preprocessing steps to ensure data quality and consistency. First, the dataset was filtered to include only entries from Pennsylvania. Missing or invalid values were removed to prevent errors during computation. Additionally, only relevant features such as restaurant name, city, star rating, and review count were selected for analysis.

Feature engineering played a crucial role in improving the effectiveness of the ranking model. A custom scoring function was developed to combine multiple attributes into a single value. While

star ratings reflect customer satisfaction, review count reflects popularity and reliability. However, review counts are often highly skewed, with a small number of restaurants having extremely large values. To address this, a logarithmic transformation was applied to the review count, reducing its dominance while preserving its importance.

The final scoring formula is defined as:

$$\text{score} = 0.7 \times \text{stars} + 0.3 \times \log(\text{review\_count} + 1)$$

This formulation ensures that both quality and popularity contribute meaningfully to the final ranking. The weighting factors (0.7 and 0.3) were chosen to prioritize customer satisfaction while still considering the credibility provided by a larger number of reviews.

## 5. Model Development

Instead of using a traditional supervised learning algorithm, this project employs a feature-based ranking model. The model takes numerical inputs such as star ratings and review counts and produces a continuous score that can be used to rank restaurants.

The input features to the model are:

- Star rating (stars)
- Review count (review\_count)

The output of the model is a computed score that represents the overall quality of a restaurant. This score is then used to sort restaurants in descending order, producing a ranked list.

The model uses a weighted linear combination of features, along with a logarithmic transformation to normalize the scale of review counts. This approach is computationally efficient and interpretable, making it suitable for real-time applications.

## Results and Evaluation

The ranking model produced intuitive and realistic results. Restaurants with high ratings and a substantial number of reviews consistently appeared at the top of the list. The model effectively distinguishes between:

- Restaurants with high ratings but few reviews (less reliable)
- Restaurants with many reviews but lower ratings (less favorable)

By balancing these factors, the model provides a more accurate representation of overall restaurant quality. The use of logarithmic scaling ensures fairness and prevents extremely popular restaurants from dominating the rankings solely due to volume.

## 6. Showcase

The final system is implemented as a full-stack web application consisting of a React frontend and a Flask backend. The frontend provides a user-friendly interface where users can select a city and view the top-ranked restaurants. The backend processes user requests, computes ranking scores, and returns the results in real time.

### System Workflow

1. The user selects a city from a dropdown menu
2. The frontend sends a request to the Flask backend API
3. The backend filters the dataset based on the selected city
4. The ranking model computes scores for each restaurant
5. The top results are returned to the frontend
6. The frontend displays the ranked list dynamically

This interaction demonstrates how the model is integrated into a real application and used for inference in real time.

## 7. Discussion and Conclusions

This project demonstrates how data-driven methods can be applied to solve practical problems in everyday life. By combining multiple features into a ranking model, the system provides more meaningful recommendations than traditional single-metric approaches.

One of the main challenges encountered was the lack of textual review data, which limited the ability to apply natural language processing techniques such as sentiment analysis. Additionally, some cities had fewer data points, which affected the availability of results for certain locations.

Despite these limitations, the project successfully demonstrates the application of machine learning concepts, including feature engineering, model design, and system integration. Future improvements could include incorporating review text for sentiment analysis, expanding the dataset to include more regions, and experimenting with more advanced machine learning models.

## 8. AI Usage

We used ChatGPT to assist with debugging integration issues between the frontend and backend, as well as to help identify and resolve coding errors encountered during development.

## 9. Overall Quality

This project demonstrates a complete end-to-end pipeline, from data collection and preprocessing to model development and deployment within a web application. It integrates concepts from machine learning, data analysis, and software engineering into a cohesive and functional system.

The report provides a detailed explanation of each stage of the project, supported by clear reasoning and structured presentation. Overall, the project reflects a strong understanding of course concepts and their application to real-world problems.